

Optimizing Greenhouse Crop Yield in the Cauvery Delta Region: Integrating PAR, pH, and Environmental Parameters Using Machine Learning

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ABSTRACT- *The fertile Cauvery Delta in Tamil Nadu contributes much to the agriculture-driven economy of India through herbaceous and horticultural crops. Traditional methods for improving crop yield in the region are based on parameters like soil moisture, nutrients, temperature, rainfall, and soil type. This study emphasized the role of Photosynthetic Active Radiation (PAR) and pH in crop yield, with PAR being an important but underused factor due to a lack of data. Through the use of Linear Regression, the relationship between PAR, meteorological parameters, temperature, and soil nutrients influenced by pH was analyzed. A Random Forest regressor modeled the interaction between PAR, Temperature, Illuminance, and Tropical Storm Risk (TSR). Six months of real-time PAR data from the weather stations in the Cauvery Delta trained the machine learning models to predict PAR and pH for the crop data sets from the Soil Testing Laboratory, Aduthurai, and Greenhouse Crops. Predictions were contrasted with temperate PAR-Nutrients-pH(TP-NP) mapping logic and optimized for yield enhancement. The R^2 statistic for PAR prediction was 0.99 for Temperature, Illuminance, and TSR and 0.79 for PAR vs. temperature. The prediction model for pH resulted in an R^2 value of 0.8. Results define optimal conditions for agricultural practices in controlled environments greenhouse crops, enhancing sustainable.*

Keywords-Greenhouse crop optimization, PAR, pH prediction, TSR, sustainable agriculture.

I. INTRODUCTION

The Cauvery delta region located in Tamil Nadu is known as the granary of the state [1] because it has rich alluvial soil and abundant water resources for

optimal plant growth. This region contributes 30 percent of the food in the state, but the numbers have been reduced to 20% due to changes in climate observed in recent years. This reduced agricultural productivity is the most important concern for smart agricultural practices. Greenhouse agriculture offers a controlled environment which includes factors like temperature, humidity, light can be manipulated to achieve maximum plant growth. This provides crop production even in times of vulnerable climatic conditions.

Among the many factors which contribute to plant growth, PAR and soil pH are two of the most crucial factors. PAR refers to the portion of sunlight that plants use for the process photosynthesis which converts light energy to chemical energy to use for food production. In greenhouse agriculture, controlling PAR, regardless of weather conditions guarantees that plants receive adequate energy for photosynthesis [2]. This reduces farmer's dependency on weather conditions for crop yield. In addition to PAR, soil pH is equally contributing to plant growth. Soil pH affects nutrient availability and determines whether the nutrients can be absorbed by plants. When the pH is too low, it creates an alkaline environment, certain nutrients cannot be absorbed by plants in an alkaline environment [3]. By carefully selecting pH within the optimal pH range of Cauvery

delta region crops and regulating PAR along with it will maximize crop yield.

To understand better the relationship between PAR and pH values in greenhouse cultivation, Table 1 show the optimal ranges of PAR and pH required for tomato [4].

TABLE I. OPTIMAL RANGES OF PAR AND PH FOR TOMATO

pH Range / Growth Suitability	Optimal PAR ($\mu\text{mol}/\text{m}^2/\text{s}$) / About Crop
0-4	Unsuitable / Not applicable / Extreme acidity; toxic for plant growth.
4-5	Marginally suitable / 400-500 / Improved nutrient retention, water holding capacity, and reduced stress effects.
5-6	Suitable / 400-600 / Water uptake, root development, and soil aeration are enhanced for improving nutrient availability.
6-7	Ideal / 500-700 / Optimal pH for nutrient absorption; promoting strong growth.
7-8	Suitable / 500-600 / Minor alkalinity impacts, supporting healthy growth.
8-10	Marginally suitable / 400-500 / Nutrient uptake is limited under saline stress conditions.
10-14	Unsuitable / Not applicable / Highly alkaline and toxic for growth.

From the Table I, the optimal pH and PAR for tomato are as follows: For tomato, well-suited pH range is 5.5-6.5 and PAR range is 400-600 $\mu\text{mol}/\text{m}^2/\text{s}$. These optimal conditions foster plant health, nutrient absorption and yield [5].

In the crop growing scenario of Cauvery delta, both linear regression and random forest regression can be employed for understanding the impact of environmental factors such as PAR and soil pH on the yield. These Machine learning algorithms are assisting in forecasting and discovering the best environment suitable for plant growing within a greenhouse because of the necessity of accuracy. Given current climate change tendencies, it is critical to use data analysis and assessment approaches to improve the utilization of the obtainable resources like PAR and pH to support and boost crop yields [6]. linear regression explains how PAR and soil pH affect crop yield, disclosing the strength and direction of their influence. Such insight makes it easier to predict yields under different conditions. Random forest regression, on the other hand, might be able to

capture more complex relationships among temperature, humidity, and light and provide improved accuracy. The combination of these models would achieve better decision-making for greenhouse gardening by tuning PAR and pH for better yields, supporting sustainable farming in the Cauvery delta region.

II. LITERATURE REVIEW

The study uses modeling techniques such as Linear Regression and Random Forest Regression to examine the relationship between plant growth in the Cauvery delta's greenhouse environment and important environmental factors, including PAR, temperature, pH, and soil nutrient composition. The objectives are focused on manipulating these variables, more particularly the PAR and pH to enhance the germination of plants and enhance the practice of sustainable agriculture in the region. The research works presented in this section are the key to the achievement of these objectives as these seek to establish the effects of environmental parameters such as PAR, pH and soil characteristics on plant health and growth.

[7] and [8] deal with the application of machine learning for modeling estimations of parameters, which depend on PAR and soil characteristics, including chlorophyll content and crop biomass. [9] and [10] present the study on the estimation of fraction of PAR absorbed by plants using spectral indices and remote sensing which give instantaneous updated information on efficiency of light absorption by the plants in your greenhouses. The pattern of how soil pH influences the growth of plants, discussed in [11] and [12]. These studies demonstrate how growth media pH affects the nutrients available.

The research work presented in [13] examines how prismatic density affects the effectiveness of light capture, another aspect of successful PAR for photo phytic processes. [14] and [15] discuss how, based on data, remote sensing intentions and a machine learning algorithm, the growth of the crops can be predicted, and the soil fertility can be monitored. These studies stress on how growth conditions can be controlled through environmental monitoring. In [16] and [13] authors used multispectral data and

machine learning to model some plant growth descriptors such as the LAI. With the use of these techniques, models will come up with better predictions on the state of plant health and growth and therefore contribute positively to determining the right conditions to foster sustainable agriculture.

In brief, the above research papers present valuable information for machine learning and remote sensing techniques and environmental monitoring systems as being useful for achieving the goal of enhancing PAR and pH for improved plant productivity. With these methodologies, there is the ability to design better strategies of managing the conditions in the green house in the Cauvery delta region and enhance the spread of efficient methods in agriculture

III. OBJECTIVE OF THE RESEARCH WORK

The primary goal of this research work is to optimize crop yield in the Cauvery Delta region by integrating PAR, pH levels, and other environmental parameters into predictive models. The research focuses on the development of region-specific insights for greenhouse crop management and enhances sustainable agriculture practices. The main contributions of the proposed work are the following:

- To analyze the relationships between PAR, temperature, soil nutrient concentrations, and pH levels using machine learning models, focusing on the Cauvery delta region.
- To use Linear Regression and Random Forest regressors to estimate crop PAR and pH values using real-time meteorological data and soil test findings.
- To use expected PAR and pH values and compare them with global standards to optimize greenhouse environmental conditions, increase crop yield and encouraging sustainable agriculture.

IV. PROPOSED METHODOLOGY

The overall architecture is divided into three phases namely:

- Integrated Soil, Crop, and Weather Data Collection Framework
- Predictive Modelling with TP-NP Mapping Logic
- Greenhouse Customizing Control (GCC) Technique

A. Stage 1: Integrated Soil, Crop, and Weather Data Collection Framework

To implement the proposed model architecture, the respective dataset required for collected from the Public Works Department (PWD). Initial weather data, specifically for PAR, was obtained from real time weather station installed at MHI sponsored TBI at SASTRA Deemed university. The weather data utilized spans the period from March to September 2024. Subsequently, the soil dataset and crop dataset were acquired from Soil testing laboratory, Aduthurai. The attributes of the datasets are detailed below:

The weather dataset comprises 17 attributes and spans data for 6 months. Each day includes 143 data points, resulting in a total dataset size of $[26,030 \times 17]$.

TABLE II. WEATHER DATASET

S.NO	WEATHER ATTRIBUTES
1	Time
2	CO2
3	Humidity
4	Illuminance
5	PAR
6	Temperature
7	Total Solar Radiation (TSR)
8	Pressure
9	Wind Speed
10	Rain Accumulation
11	PPM10
12	PPM2_5
13	Wind Angle
14	Wind Speed Level
15	Maximum Rainfall
16	Wind Direction

The soil fertility dataset is composed of soil parameters for various blocks and villages of Thanjavur district. There are 14 blocks, and each block's soil fertility index namely micro (Zn, B, Fe, Mn and Cu) and macro nutrients (N, P, K, S) constitution is available in the dataset

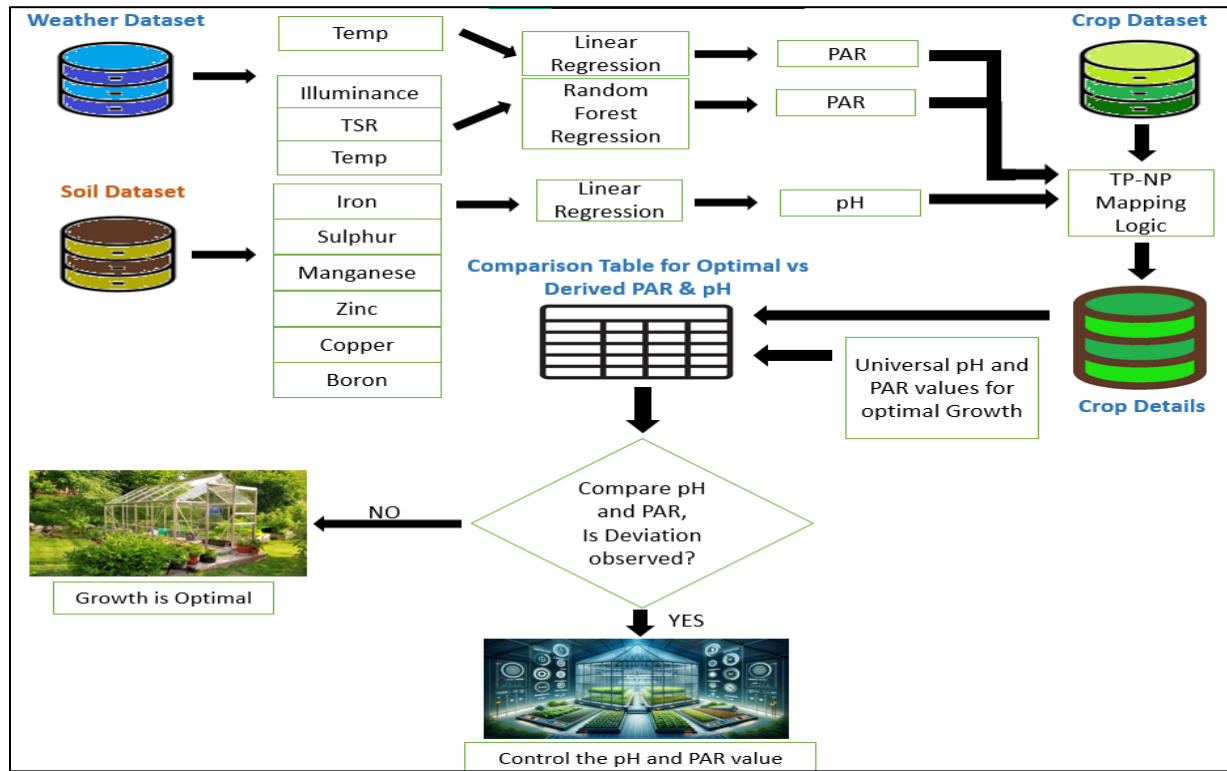


Fig. 1. PROPOSED SYSTEM ARCHITECTURE

TABLE III. SOIL DATASET ATTRIBUTES

S.NO	WEATHER ATTRIBUTES
1	BLOCK
2	Lime Status
3	Texture
4	pH
5	Electrical Conductivity (EC)
6	Nitrogen (N)
7	Phosphorous (P)
8	Potassium (K)
9	Sulphur (S)
10	Zinc (Zn)
11	Iron (Fe)
12	Copper (Cu)
13	Manganese (Mn)
14	Boron (B)

B. Stage 2: Predictive Modelling with TP-NP Mapping Logic

Predictive modelling was conducted using the TP-NP Logic Crop Dataset to extract crop details for Stage 3, facilitating greenhouse control. During this stage, a Linear Regression Model was constructed for Temperature vs. PAR, while a Random Forest Regression Model was employed for Temperature, Illuminance, & TSR and PAR. The TP component involved the prediction of PAR based on temperature using the weather dataset model, integrated into the crop dataset via its average temperature feature to compute the average PAR. Similarly, the Nutrients model, developed using Linear Regression and

Dataset, was utilized in the crop dataset's nutrient attributes to estimate the pH value.

$$PAR = 2.97 * Temp + 479.82 \quad (1)$$

From the block wise fertility index in soil dataset, it is identified that only B, Mn, Cu, Fe, Zn and S exhibit a high correlation with pH. Using these variables exclusively, pH is modelled through Linear Regression and derived the following equation to map nutrients to pH.

$$pH = 7.0546 + (-0.4087) * B + (-0.3675) * Mn + (-1.3228) * Cu + (0.2002) * Fe + (1.0895) * Zn + (0.0346) * S \quad (2)$$

The Crop dataset Attributes is composed of 47 different crops grown in Thanjavur district

TABLE IV. CROP DATASET ATTRIBUTES

S.NO	WEATHER ATTRIBUTES
1	Crop
2	Average Temperature
3	Average Rainfall
4	Rainfall Contribution (%)
5	Relative Humidity
6	Season Classification
7	Nitrogen (N)
8	Phosphorous (P)
9	Potassium (K)
10	Sulphur (S)
11	Zinc (Zn)
12	Boron (B)
13	Iron (Fe)
14	Manganese (Mn)
15	Copper (Cu)

C. Stage 3: Greenhouse Customizing Control (GCC) Technique

From stage 2, the system reads the specific pH and PAR values for the crops from the crop details. This value is compared to a globally established ideal growth standard through a comparison table. Provided that the values are acceptable, the system determines that growth conditions are adequate to produce yields above the average, higher growth and no action is needed. However, if any discrepancy arises, the system changes and fine-tunes the regulating mechanism to provide an adequate environment for the crops such that they remain healthy and productive throughout their life span.

For pH imbalances, basic fertilizer administered if the pH is below the desired base level, and acidic fertilizer is used if the pH is too high. The system also adjusts the PAR value if it is above the ideal range. A shielding layer is introduced to reduce it. However, if the PAR value is too low, the system provides artificial lighting simulating solar radiation. By this dynamic control of key factors, the system ensures that the conditions inside the greenhouse are always ideal for the crop, hence maximizing growth and yield efficiency.

IV. RESULTS AND DISCUSSION

Relationship between Photosynthetically Active Radiation and Key Environmental Variables

A. a.. PAR and Temperature

Due to the shared dependence on solar radiation, both PAR and temperature are strongly correlated. Peak solar hours lead to simultaneous increases in temperature and PAR, even though the relationship weakens under cloud cover, where PAR drops more sharply. With higher latitudes and colder seasons showing greater variability, seasonal and geographical factors further modulate this interplay. Therefore, understanding this relationship is critical for optimizing crop productivity and photosynthesis in disaster preparedness and regional climate modeling.

b. PAR and Tropical Storm Risk (TSR)

TSR is influenced by sea surface temperatures, which drives the oceanic heat energy. It also shows a linear relationship with TSR under stable climatic conditions, since PAR is a subset of solar radiation

Monitoring PAR and TSR together provides insights into TSR which aids understanding this relationship is critical for optimizing crop productivity and photosynthesis

c. PAR and Illuminance

As both originate from the visible light spectrum (400-700 nm), PAR and Illuminance are closely linked. Illuminance sensors can act as cost-effective proxies for PAR in controlled environments. This correlation is especially useful for greenhouse lighting optimization, even though direct PAR measurements are preferred for precision applications.

d Model Performance and Forecasting

The PAR analysis integrates linear regression models for parameter predictions. The models R^2 values for predicting PAR using temperature, TSR, and Illuminance is 0.99, which reflects a strong predictive accuracy using Random forest regressor and for predicting PAR using Temperature is 0.79 using linear regression.

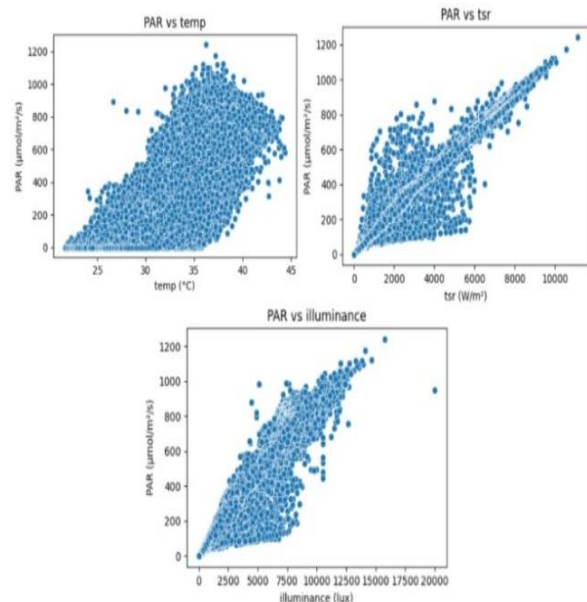


Fig 2 COMPARATIVE ANALYSIS OF PAR AND WEATHER PARAMETERS

B. Relationship between Soil pH and Nutrients

Soil pH plays a very important role in determining the availability and uptake of various nutrients, both macronutrients (N,P,K,S) and micronutrients (Zn,B,Fe,Mn,Cu). This influences chemical solubility and biological processes in the soil.

a. The following nutrients are highly related to pH

- Iron (Fe): Decreases sharply in alkaline soils due to precipitation and mostly available in acidic soils (pH 4.0–6.5).
- Manganese (Mn): It forms insoluble oxides in alkaline conditions but soluble in acidic soils (pH 5.0–6.5).

b. The following nutrients are moderately related to pH

- Nitrogen (N): Its availability peaks between pH range 6.0–8.0. Alkaline conditions promote nitrification, but efficiency is reduced due to extreme pH levels through volatilization or denitrification.
- Sulfur (S): More mobile and available in acidic soils and less so in alkaline conditions.
- Zinc (Zn): Due to precipitation, solubility decreases in alkaline soils (above pH 7.0), even though it remains moderately influenced compared to other micronutrients.

c. The following nutrients are weakly related to pH

- Boron (B): Availability decreases slightly in both highly alkaline and highly acidic soils, but it is less influenced by pH than others.
- Copper (Cu): Even though it is less pH-sensitive than phosphorus or manganese, availability reduces in alkaline soils (above pH 6.5).

C. Greenhouse Crop Cultivation: Optimal PAR and pH Analysis

PAR and pH are crucial environmental factors influencing greenhouse crop productivity. This is derived from analysis evaluates dataset values against few optimal cultivation parameters derived from scientific research for key crops, including tomatoes, cucumbers, green chilies, onions, and lady's fingers with the predictive model. The results highlights alignment and areas for improvement which provides actionable insights for enhanced greenhouse management practices.

To address the slight pH deviation for cucumber, the following solutions can be applied to it:

1. Acidic Amendments: Adding sulfur or aluminum sulfate to gradually lower soil pH
2. Organic Matter: Incorporating peat moss or composted manure to naturally acidify the soil.
3. Acidic Fertilizers: promoting soil acidity using ammonium sulfate-based fertilizers.
4. Water Management: Using phosphoric or citric acid, acidify alkaline irrigation water.
5. Mulching: Applying acidic mulches like pine needles or oak leaves.

Regular monitoring and adjustments made will ensure pH aligns with the optimal range.

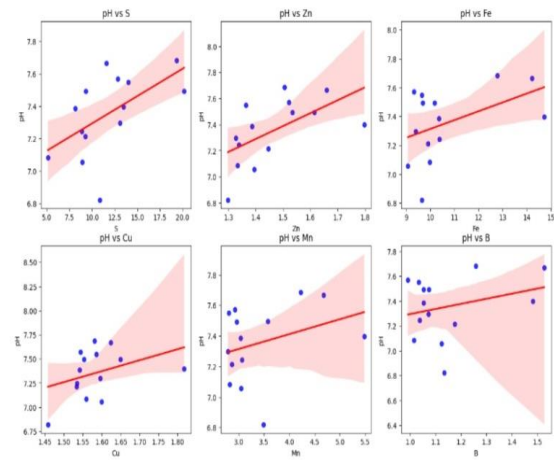


Fig. 3 COMPARATIVE ANALYSIS OF SOIL pH AND NUTRIENTS

TABLE V. COMPARISON OF OPTIMAL PAR AND PH FOR CROPS

Crop	Dataset PAR ($\mu\text{mol}/\text{m}^2/\text{s}$)	Optimal PAR Range ($\mu\text{mol}/\text{m}^2/\text{s}$)	Dataset pH	Optimal pH Range	PAR Alignment
Tomato	489.65 $\mu\text{mol}/\text{m}^2/\text{s}$	400-600 $\mu\text{mol}/\text{m}^2/\text{s}$	6.0	6.0 - 6.8	Excellent match
Cucumber	498.77 $\mu\text{mol}/\text{m}^2/\text{s}$	500-600 $\mu\text{mol}/\text{m}^2/\text{s}$	6.75	6.0 - 6.8	Very Close
Green Chilli	489.65 $\mu\text{mol}/\text{m}^2/\text{s}$	400-550 $\mu\text{mol}/\text{m}^2/\text{s}$	6.15	6.0 - 7.0	Excellent match
Onion	459.23 $\mu\text{mol}/\text{m}^2/\text{s}$	350-500 $\mu\text{mol}/\text{m}^2/\text{s}$	6.5	5.8 - 6.5	Excellent match
Ladies Finger	498.77 $\mu\text{mol}/\text{m}^2/\text{s}$	400-600 $\mu\text{mol}/\text{m}^2/\text{s}$	6.25	6.0 - 6.8	Excellent match

VI.CONCLUSION

The research work presented in this paper highlights the crucial role of integrating Photosynthetic Active Radiation (PAR) and pH factors to enhance the yield of greenhouse crops/plants in the Cauvery delta region. Using Machine Learning models like linear regression and random forest regression, the study establishes a strong correlation between environmental factors such as PAR, temperature, illuminance, total solar radiation & soil factors such as pH, nutrients affecting the yield. The major findings include achieving a high R^2 score of 0.99 for PAR prediction when incorporating temperature, Illuminance, and solar radiation, and 0.8 for pH prediction, underscoring its importance in nutrient management for the greenhouse crops/plants. Also, this paper emphasizes the need for TP-NP mapping and AI-driven forecasting to monitor greenhouse conditions effectively, advocating for aligning local PAR and pH standards with global benchmarks. This approach fosters sustainable agriculture while addressing the challenges of adaptability and productivity in the region.

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